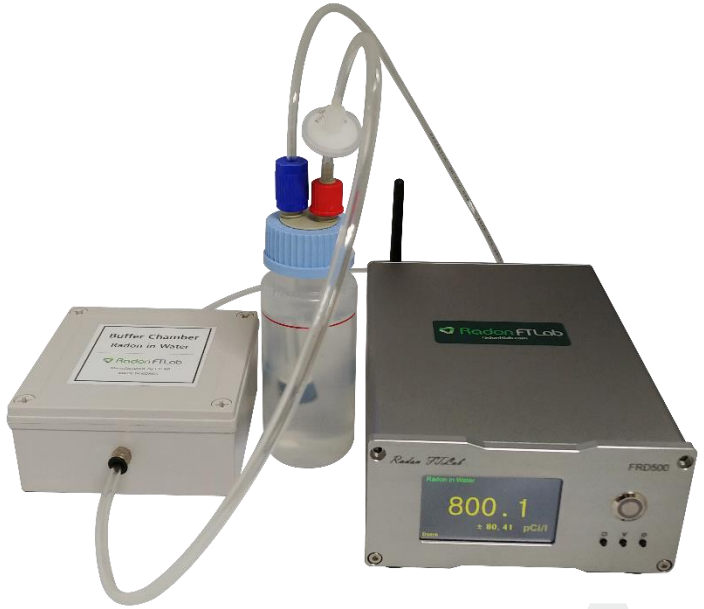
FRD500 Radon in Water Quick Manual (User's Manual)

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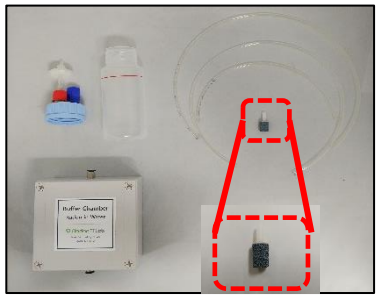


※ This device is an electromagnetic compatible device for business use(Class A)that the seller or the user should be aware of, and is intended for use besides home.

FRD500 Radon in Water Kit Quick Manual

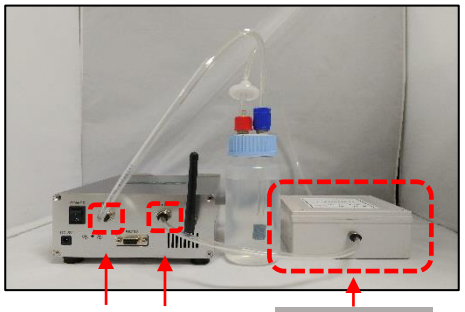
➢ Connection and Basic Sequence of Operations

[1]



Components

[3]



Outlet | Inlet | Buffer Chamber

[2]

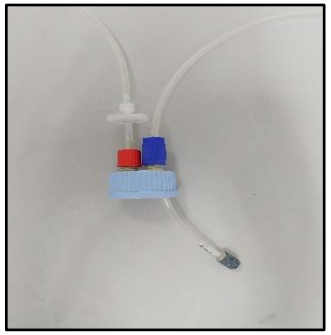
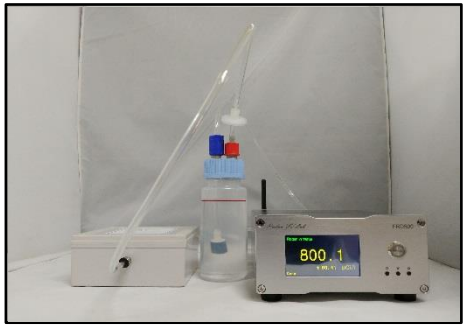


Figure of water bottle lid connection

[4]



Water Kit connected to FRD500

1. FRD500 Radon in Water Kit consists of a water bottle, lid connection, Bubble Stone, Buffer Chamber, and a hose, as shown in [1].
2. Assemble the hose on the lid of the water bottle s shown in [2]. Cut the hose by 60cm long and connect it to the red connector dust filter, and also insert a 60cm long hose into the blue connector and tighten the connector by hand securely. At this time, the Bubble Stone must be inserted into the hose that entered the water bottle, so adjust the location of the hose properly so that the Bubble Stone does not touch the bottom of the water bottle.
3. Insert the hose of the blue connector into the outlet on the back of the FRD500 and connect the hose of the red connector to the Buffer Chamber as shown in [3]. In the opposite side of the Buffer Chamber, cut the hose by 30cm long and connect to the Inlet on the back of the FRD500. Check the connections several times to make sure there are no leaks. Buffer Chamber has no distinction between the front and the back, so connect it however you want.
4. Before starting the measurement, connect the Buffer Chamber and the FRD500 in a well ventilated area and operate them in Flushing mode for at least 10 minutes before performing the measurement. In particular, it is strongly recommended to flow for at least 20 minutes after measuring high concentrations of radon (including airborne and underwater radon). When measuring again immediately after measuring a high concentration of radon, Flushing mode must be used, as the remaining radon at the Buffer Chamber and the measurement section interfere with the new measurement

➢ How to Measure

1. Connect the Antenna to the Antenna Port on the back of the FRD500.
2. Connect the DC 5V/2A adaptor provided with the DC 5V power connector or the auxiliary battery having a higher power, and turn on the Power Switch on the back.
3. Select Radon in Water mode using V Button on the measurement mode selection screen. If it is disabled in dark gray, connect the smartphone app to activate Radon in Water in Config Setting on the upper right side. It can also be activated in the Parameter Setup in the PC program.
4. Fill the Water Bottle with the water to be measured and firmly close the lid.
5. After checking that there are no leaks in the hose connection, press Start/Stop Button to start the measurement. The measurement order is; 10 minutes pumping - 5 minutes waiting - 10 minutes measurement 4 times, with the total of 55 minutes. By the third measurement, the display will show the result, and after the fourth measurement, the average of the 4 measurements will appear on the display, and the measurement will be automatically terminated.)

➢ How to Fill Water

1. Fill the bottle with water up to the red mark.
2. When filling water, fill it not shocking the water as much as possible. Physical impact on water containing radon may not make accurate measurements as radon exits the water. If possible, the impact can be reduced by filling the water through a dropping pipette and funnel.
3. The plastic bucket for measurement is a container for collecting and measuring on-site. To measure after collecting the water and move to another place, use a glass bottle, not a plastic bottle, and fill it up full and move carefully so that there is no swinging as much as possible. Also, from the point of collection, radon collapses with half-life, so the time from collection to measurement should be as short as possible.

➢ Data Storage and Check

1. The stored Log is marked with Stored Time and Radon in Water in the Log List.
2. The log consists of 4 measurements and an average value. You can check the measured value on the FRD500 device itself, or you can also use a smartphone app or PC program.
3. Use one Slot for a single measurement for the Log of Radon in Water. If there are 10 stored logs, the oldest log will be automatically deleted, so the data must be backed up.
4. You can check and save it in the app or PC program even during measurement.
   (Real-time measurements are displayed when Auto is set in PC program)

➢ Cautions

1. Make sure to check the direction of the Inlet and Outlet so that accurate measurements can be made.
2. Make sure to match the amount of water to the marked line. A smaller or larger amount of water can result in a smaller or bigger result.
3. If you use an adaptor and battery with a power of 5V/2A or less, pumping and measurement may not proceed normally.
4. If you use water that contains a lot of soil several times, it may make less bubbles due to foreign substances in Bubble Stone. Clean it with clean water or purchase a new Bubble Stone for use. If air bubbles do not develop well, the result may be a small value.
5. The measurement range of Radon in Water is 1 ~ 555,000 Bq/m3 (0.1 to 15,000 pCi/L).
6. Be careful not to let water get into the FRD500 body.